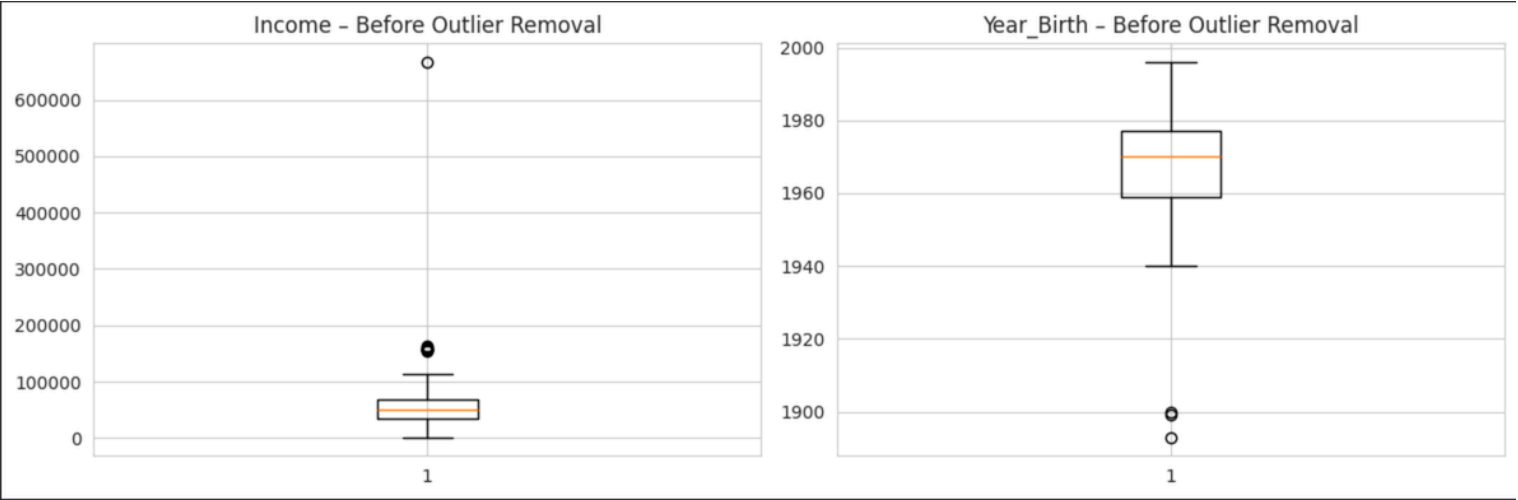


LAB EXAM part 1

Kanafina Dilnaz 535225

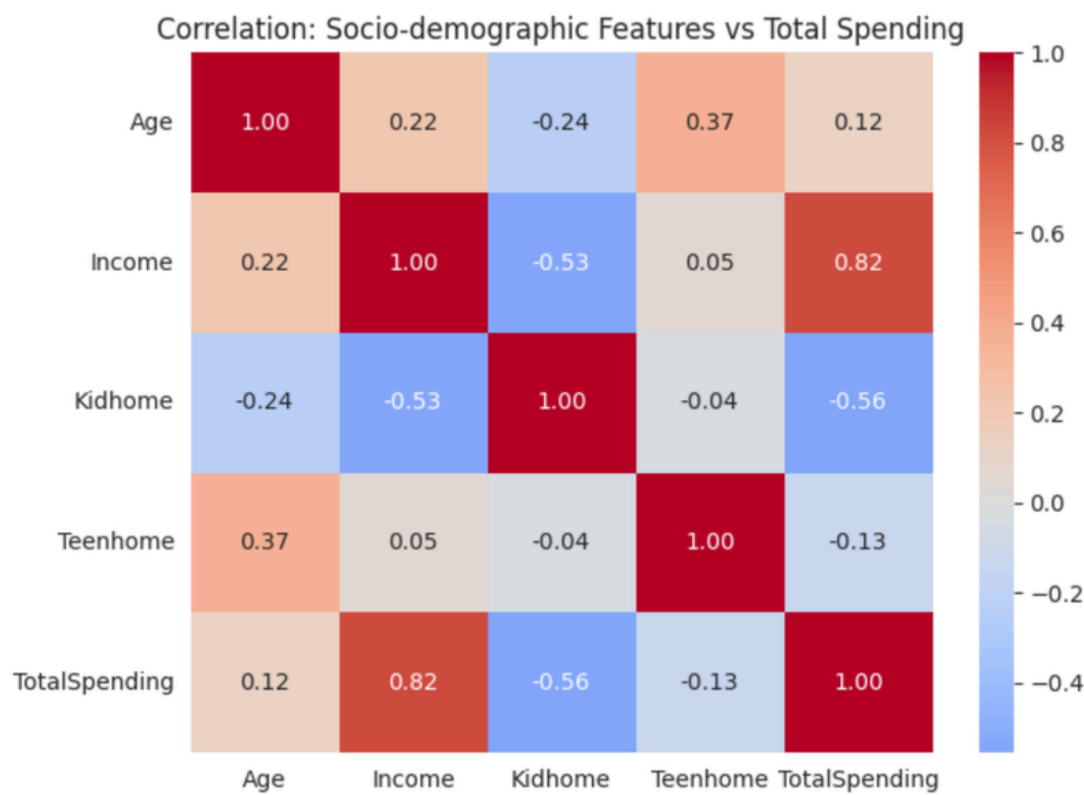
task A.

Two columns required attention. The Income column contained a value of 666,666 — over 7 times the median — which was identified as a data entry error and removed by capping at the 99th percentile. The Year_Birth column contained entries dating back to 1893, implying customers over 130 years old, which is biologically impossible. All customers born before 1924 were exclude

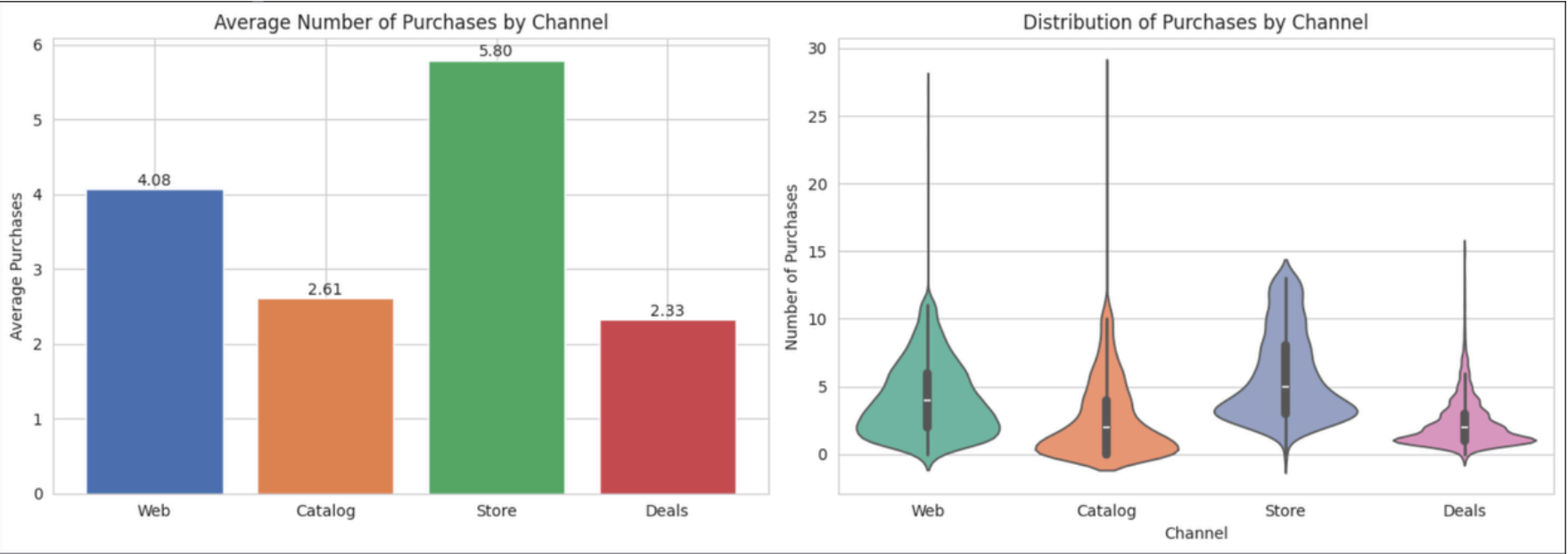


A7. Cleaning Summary			
Issue	Rows Affected	Action	Reason
Missing <code>Income</code>	24	Imputed with median	Low rate (<2%), median robust to outliers
<code>Dt_Customer</code> type	all	Converted to datetime	Required for date arithmetic
<code>Marital_Status</code> = 'Alone'	3	Mapped to 'Single'	Semantically identical
<code>Marital_Status</code> = 'Absurd'/'YOLO'	4	Dropped	Invalid entries
<code>Income</code> = 666,666	1	Dropped (>99th pct)	Data entry error
<code>Year_Birth</code> < 1924	3	Dropped	Implausible age (>100 years)
BOM in column name	1 col	Renamed to 'ID'	Encoding artefact

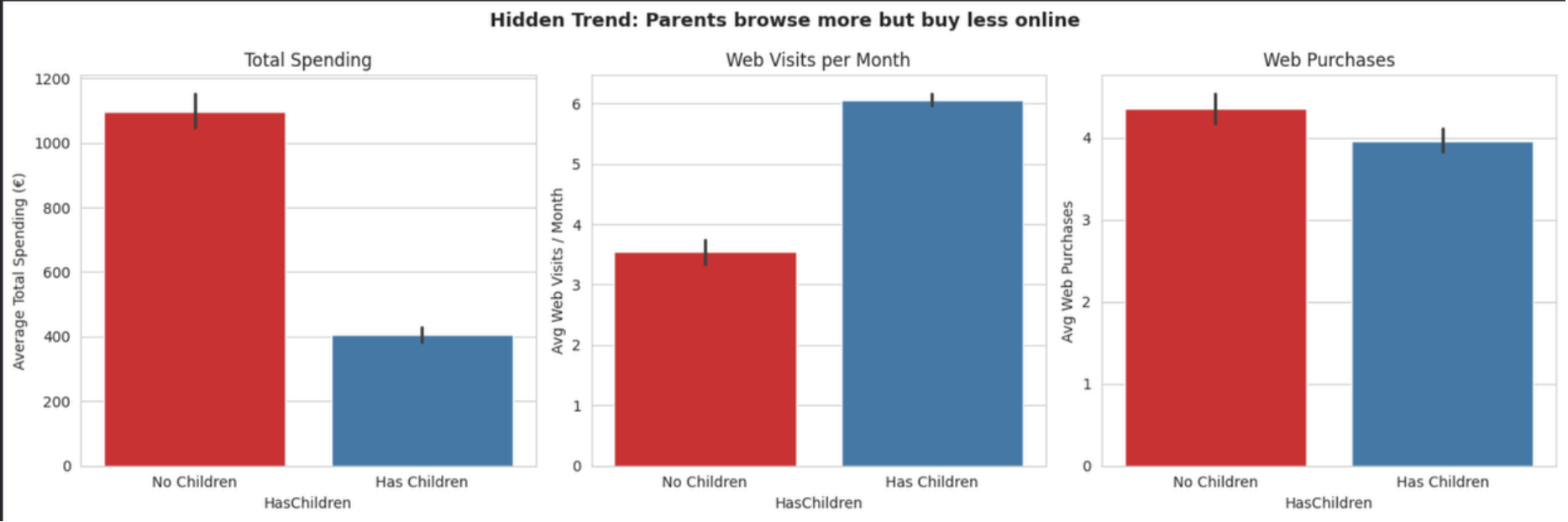
task B.



- Correlation Matrix showed us a clear results with sagging features as Kidhome and Teenhome, so individuals with children.
- Customers with high level education tend to spend more which alings with the income education link.

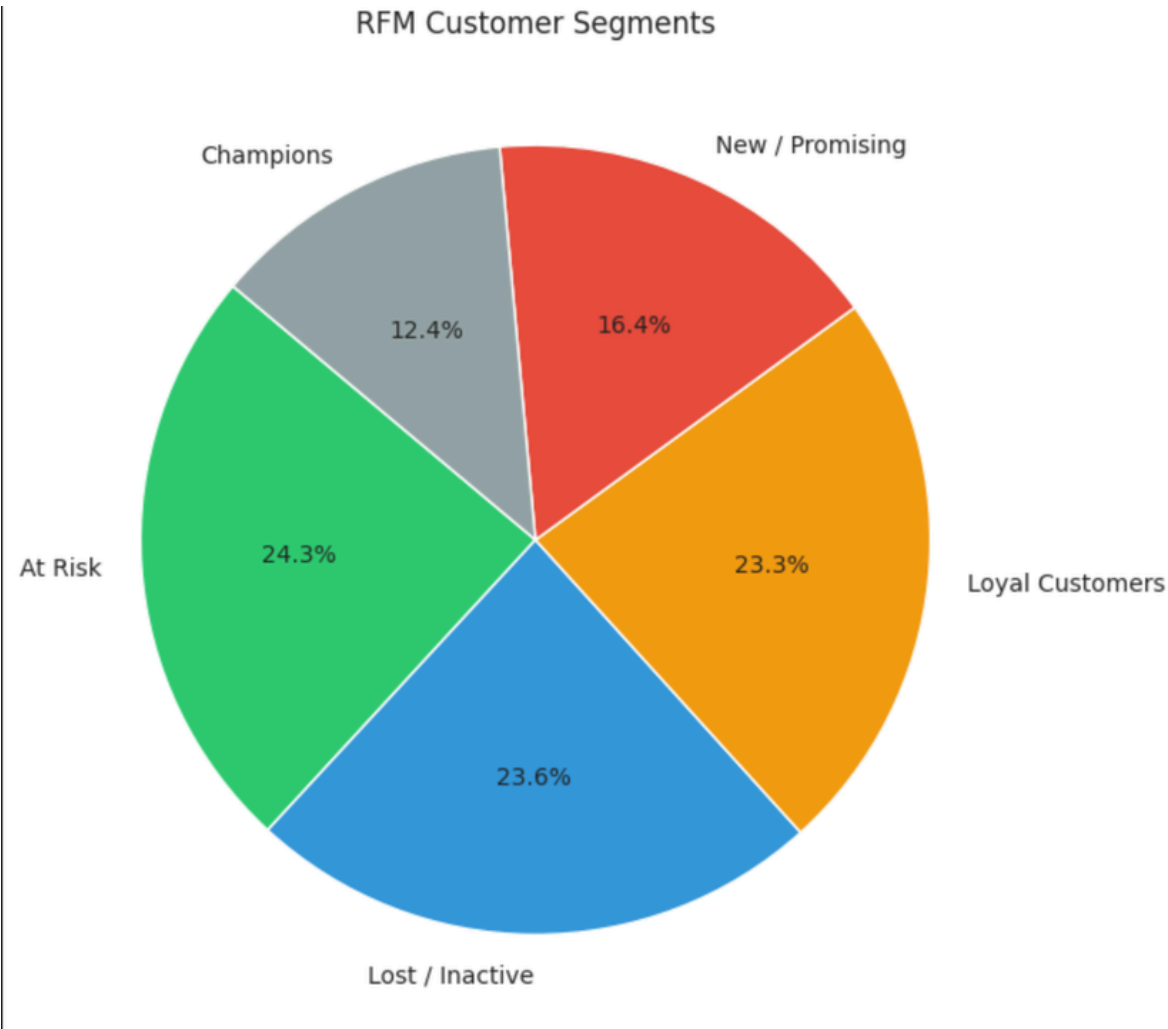


Customers still prefer physical store, which showed with significant average purchases (5.80)



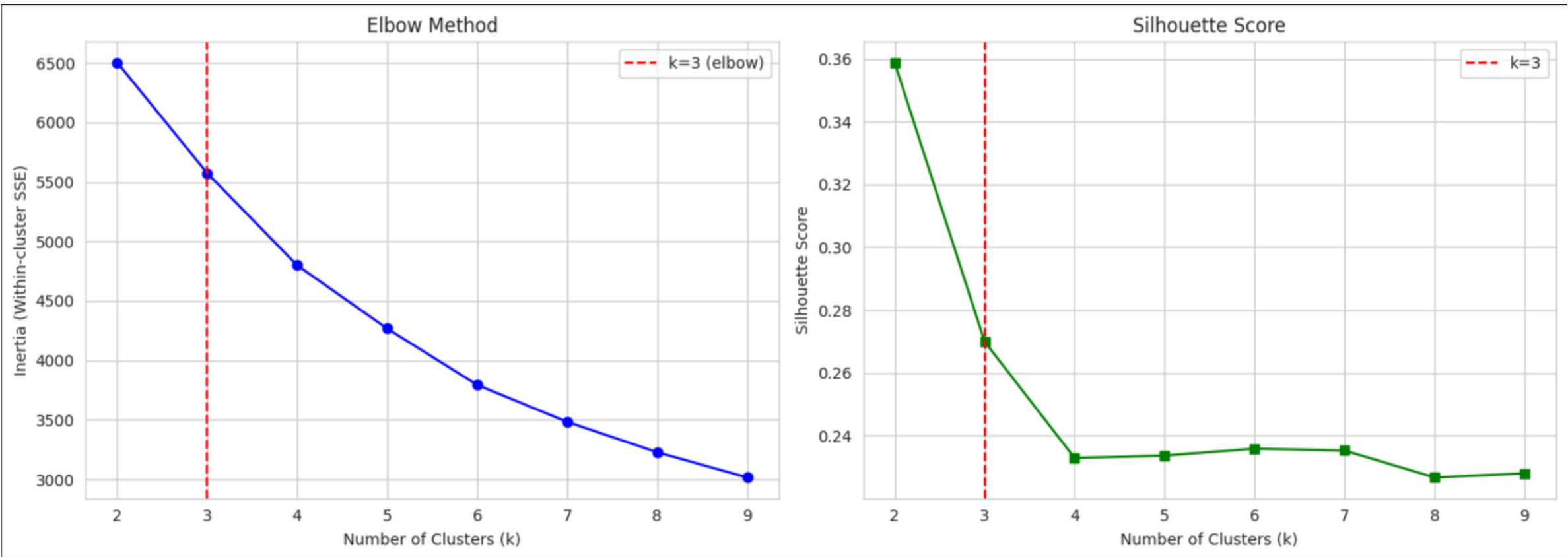
Hidden Trend to see are customers with children, there web visit more than other but this cannot be said about the purchases they make. Marketing opportunity, campaigns are aimed at clients with children.

task C.



RFM Heuristic Segmentation were used to give an estimate of 5 different segments based on recency score, frequency score, monetary score and total.

task D.



In the graphs clearly see the drop out after certain k clustering.



Having k = 3 and RFM segment we can see the difference on each segmentation. The main thing is RFM put Champions and At Risk customers in completely different segments — but K-Means put many of them in the same cluster (Cluster 1). This means they share similar demographic profiles